TulipaClustering.jl

TulipaEnergy

A Hull Clustering with Blended Representative Periods Approach for Energy System Optimization Models

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Why it works

- Selection of Extreme Points: Extreme data points for better optimization results with lower regret.
- Conic Sums for Blending: Blended RPs use conic sums to reduce the number of RPs, boost efficiency, and keep data accuracy.
- Main benefits: Improved Solution Quality and Reduction in Computational Burden

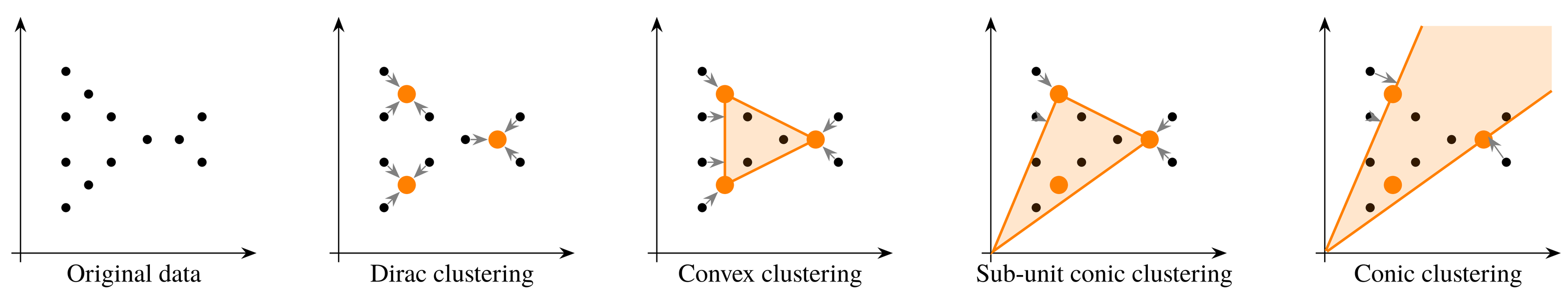


Figure 1: *Projection errors when approximating base period data using different weight types.*

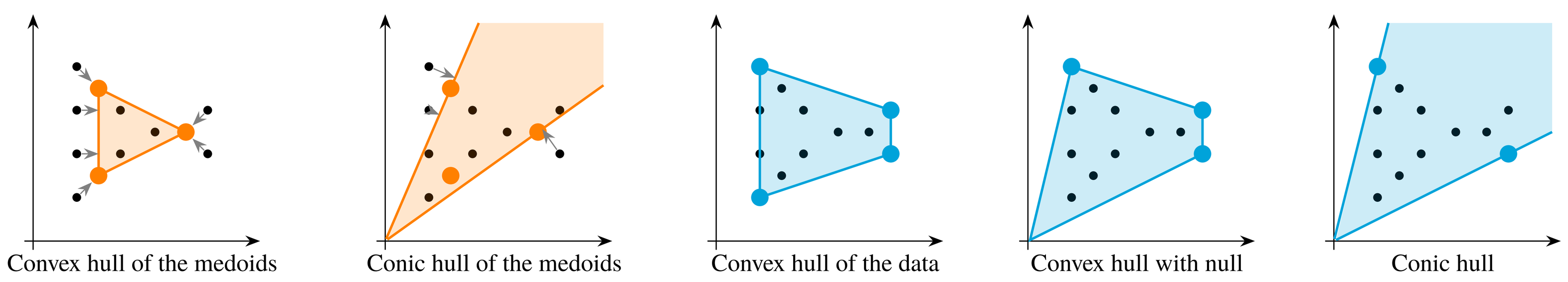
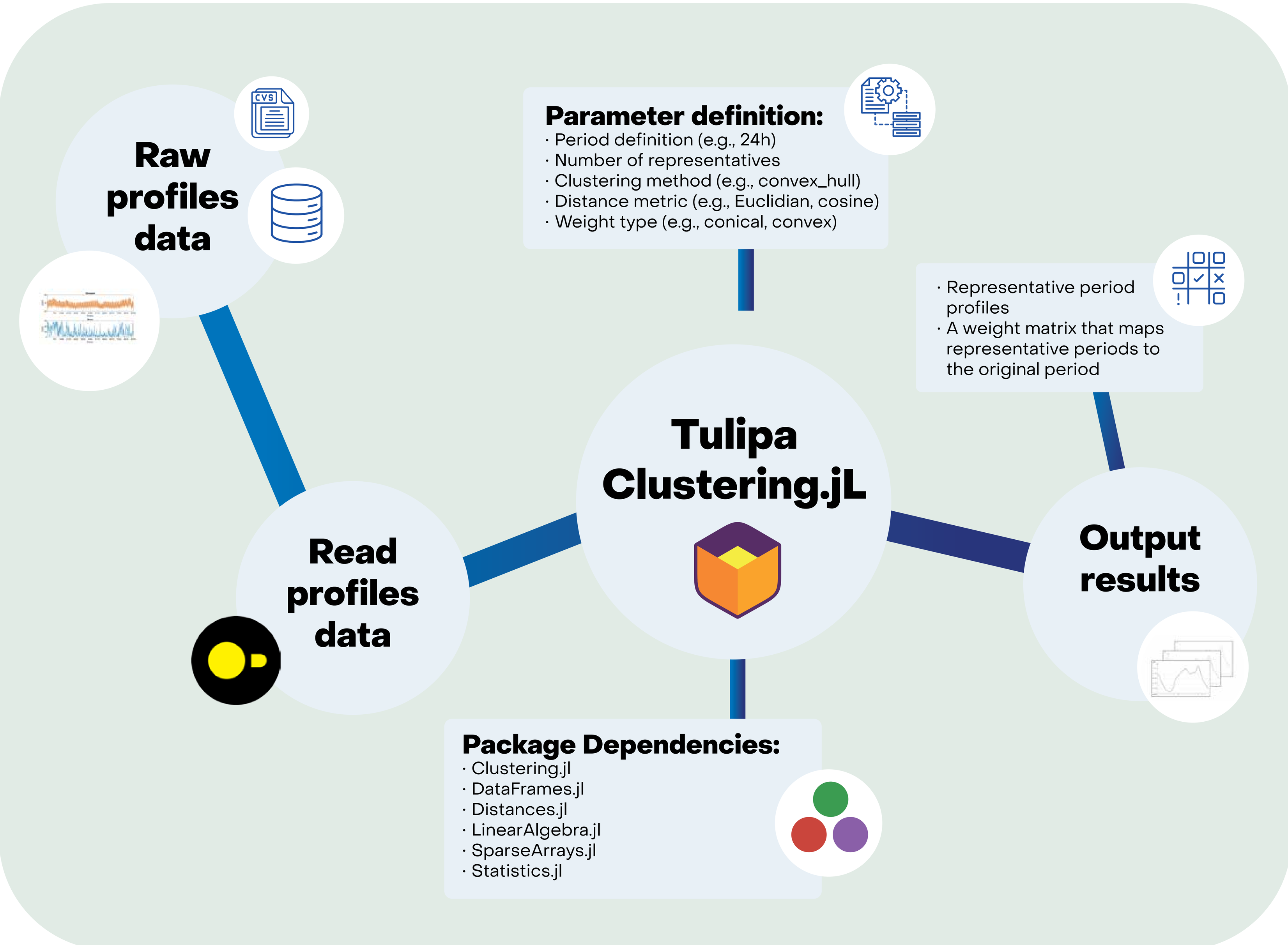


Figure 2: *Geometric interpretation of different hull types.*



Code Snippet Basic Usage

```julia
import TulipaIO as TIO
import TulipaClustering as TC
using DuckDB

# Create the database connection and read the input data
connection = DBInterface.connect(DuckDB.DB)
input_dir = "my-profiles"
TIO.read_csv_folder(connection, input_dir)
profiles = TIO.get_table(connection, "profiles_periods")

# Create the periods for clustering
period_duration = 24
TC.split_into_periods!(profiles; period_duration)

# Find representative periods and fit the weights
num_rep_periods = 12
clustering_result = TC.find_representative_periods(profiles, num_rep_periods)
TC.fit_rep_period_weights!(clustering_result)

# Write the clustering result back to the database
TC.write_clustering_result_to_tables(connection, clustering_result)
```

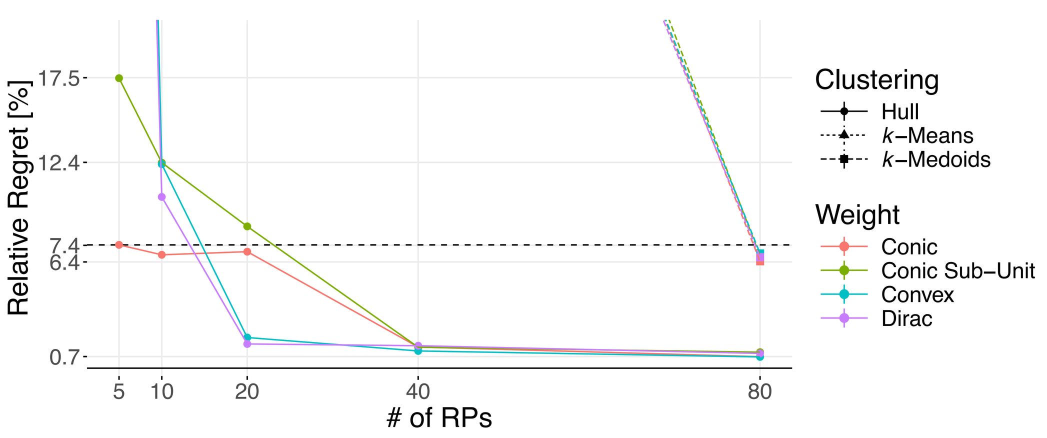


Figure 3: GEP: *Hull clustering achieves ~7.4% regret with 5 RPs, outperforming k-means/medoids.*

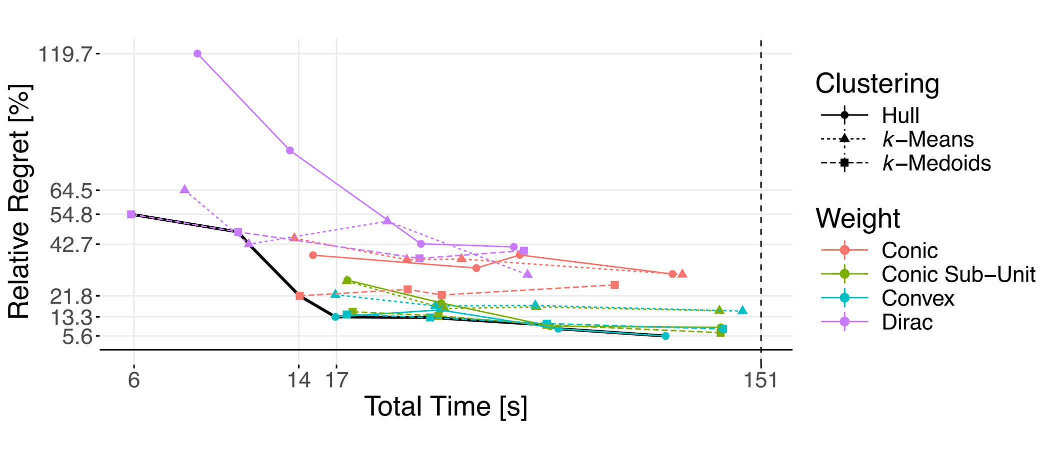


Figure 4: P2X: *Convex-weight hull sits on the Pareto frontier (<15% regret in ~17 s).*

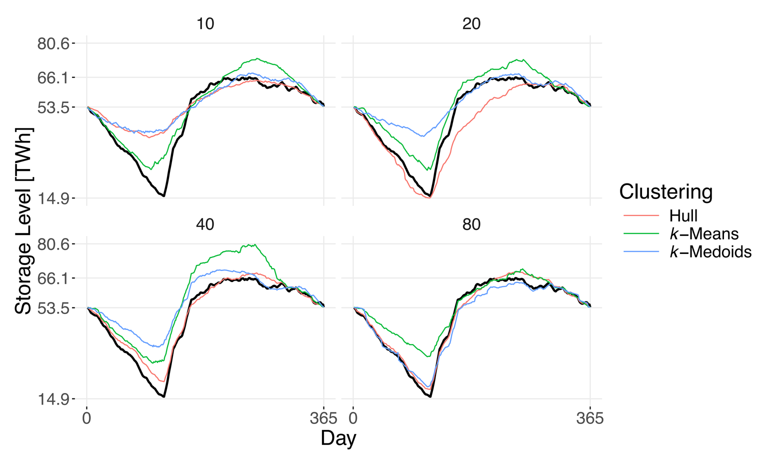


Figure 5: P2X: *Seasonal storage trajectory closely tracks the full model from ~40 RPs.*

References & Resources

1. **Paper:** Hull Clustering & Blended RPs (preprint) https://arxiv.org/abs/2508.21641
2. **Package:** TulipaClustering. https://github.com/TulipaEnergy/TulipaClustering.jl
3. **Case studies & data scripts** https://github.com/greg-neustroev/blended-rep-periods

The Blended reduces regret by 27% while using 12.5% fewer representatives compared to the Traditional

Approach	Traditional	Blended
Clustering method	K-medoids	convex hull
Weight type	dirac	convex
Num. Representatives	**80%**	**10**
Regret	**40%**	**13%**
Total time to solve (*)	44 s	6 s
Time to cluster	0.6 s	3 s

Partners

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